ZOMATO DATA ANALYSIS REPORT

Restaurant Analytics, Exploratory Data Analysis & Strategic Business Recommendations

DATASET SNAPSHOT
9,551 Outlets
15 Countries
21 Columns

Tools: Python · Pandas · Seaborn · Leaflet

Executive Summary: This report presents the end-to-end findings from an extensive **10-Phase Exploratory Data Analysis (EDA)** conducted on the global Zomato restaurant ecosystem. It provides actionable solutions across **6 Business Cases**, evaluating pricing dynamics, review reliability, service adoption (delivery and table reservations), cuisine satisfaction vs. saturation, and geographical concentration risks across Indian and international culinary hubs.

1. About Zomato & Project Scope

About Zomato:

- Founded in **2008** by **Deepinder Goyal** and **Pankaj Chaddah** in Gurugram, India.
- Primary business spans restaurant aggregation, digital menus, user ratings, online food delivery, and dining reservation services.
- Operates across 10,000+ cities globally, providing critical marketing and discovery infrastructure to restaurateurs.

About Our Project:

- Indian culinary culture is renowned for immense diversity, with rapid growth in dining out and doorstep delivery.
- **Customer Goal:** Identify highest-rated eateries, compare meal costs, and verify authentic social proof.
- **Business Goal:** Pinpoint underrepresented cities, address unrated listings, and expand high-margin delivery & booking tiers.

2. Data Preparation & Feature Schema

Feature Name	Type	Analytical Definition & Practical Meaning
NAME / Restaurant Name	Text	Official registered commercial brand name of the restaurant outlet.
PRICE / Price range	Int (1–4)	Relative economic category: 1=Budget, 2=Moderate, 3=Expensive, 4=Fine Dining.
Average Cost for two	Numeric	Estimated standard meal expense for two guests in local legal currency.
CUSINE_CATEGORY / Cuisines	Text	Multi-label comma-separated food types (e.g. North Indian, Chinese, Italian).
CITY / City	Text	Municipality or metropolitan market where the venue operates.
REGION / Locality	Text	Specific neighborhood or commercial sub-district (e.g., Connaught Place).
RATING_TYPE / Rating text	Categorical	Six qualitative evaluation tiers: Excellent, Very Good, Good, Average, Poor, Not rated.
RATING / Aggregate rating	Float (0–5)	Weighted customer rating. *0.0 strictly indicates unrated listings*.
VOTES / Votes	Numeric	Total cumulative reviews submitted by verified community diners.
Has Online delivery	Binary	Flag indicating whether doorstep delivery is operational (Yes / No).
Has Table booking	Binary	Flag indicating whether dining table reservations are enabled (Yes / No).

3. Data Integration & Pre-processing Pipeline

Key Pointwise Pre-processing Milestones:

• **Feature Pruning:** Administrative metadata lacking predictive value (URL, PAGE NO, TIMING) was excised.
• **Type Coercion:** String ratings and vote strings were coerced to numeric values via errors='coerce'.
• **Custom Null Token Cleansing:** String artifacts ('none', 'NEW', 'Not rated', 'None') were mapped to standard NaN.
• **Groupby Mean Imputation:** Missing numeric metrics were imputed using local grouping by ['CITY', 'CUSINE TYPE'].
• **Isolating 'Not Rated' Outlets:** 2,148 unrated outlets (22.5%) were encoded with an Is Rated flag so they do not falsely depress true restaurant performance (3.44★ vs 2.67★).
• **Zero Row Deletions:** 9 missing values in Cuisines were imputed as 'Unknown', preserving 100% of the 9,551 original records.

```
# Python Data Integration & Cleaning Execution Script
import pandas as pd, numpy as np, glob
all_files = glob.glob(r'Input/*.csv')
df = pd.DataFrame(np.vstack([pd.read_csv(f, delimiter='|').values for f in all_files]), columns=[...])
df.drop(columns=['PAGE NO', 'URL', 'TIMING'], inplace=True)
df['RATING'] = pd.to_numeric(df['RATING'], errors='coerce')
df['RATING'] = df['RATING'].fillna(df.groupby(['CITY', 'CUSINE TYPE'])['RATING'].transform('mean'))
```

4. Data Visualization & The 6 Business Cases

Business Case 1: Check Cuisines and Ratings in City Localities

• **Objective:** Enable customers to filter restaurants by neighborhood and cuisine with verified rating transparency.
• **Findings:** International cuisines (Italian: 3.95★, American: 3.92★, Mexican: 3.85★) command the highest customer satisfaction scores.
• **Action:** Prioritize top-rated specialized cuisine badges in regional search algorithms to increase order conversion.

Business Case 2: Average Rating vs Average Price across Regions

• **Objective:** Analyze if expensive meals guarantee superior dining satisfaction.
• **Findings:** Price Range 1 (Budget) averages 3.24★, while Price Range 4 (Fine Dining) averages 3.89★. However, correlation with raw cost is negligible (r = +0.052). High cost alone does not guarantee praise.
• **Action:** Promote budget venues with 4.0+ ratings as 'Value Champions' to build diner loyalty.

Business Case 3: Restaurant Density & Geographic Expansion

• **Objective:** Map geographic penetration and uncover regional expansion targets.
• **Findings:** Over 80.3% of listings are concentrated across New Delhi (5,473), Gurgaon (1,118), and Noida (1,080). Tier-1 tech hubs (Bangalore, Mumbai, Pune, Hyderabad) have minimal listings in this repository.
• **Action:** Deploy merchant acquisition teams into fast-growing southern and western hubs to diversify geographic risk.

Business Case 4: Restaurant Search & Votes (Social Proof Correlation)

• **Objective:** Measure how community vote counts correlate with rating stability and consumer trust.
• **Findings:** Strong positive correlation of r = +0.409 between review volume and rating. Venues with >1,000 votes rarely drop below 3.8★. Famous outliers (Toit: 10,934 votes, Truffles: 9,667 votes) serve as platform anchors.
• **Action:** Implement post-order review incentives to help newly onboarded restaurants cross the 50-vote visibility barrier.

Business Case 5: Saturated vs High-Demand Cuisines

• **Objective:** Assess supply vs. demand economics across culinary categories.
• **Findings:** North Indian cuisine is saturated (936 outlets, 3.38★ average rating), resulting in price wars. Conversely, specialized continental, cafe, and bakery offerings achieve higher margins and review scores.
• **Action:** Advise cloud kitchen entrepreneurs to fill regional cuisine deficits rather than duplicating North Indian concepts.

Business Case 6: Map Plotting & Spatial Geospatial Distribution

• **Objective:** Visualize dining density, logistics corridors, and multi-country footprints.
• **Findings:** Dining clusters hug major transport corridors. The dataset spans 15 nations (India, US, UK, UAE, Brazil, etc.), requiring local currency isolation (Indian Rupees Rs.).
• **Action:** Integrated interactive Carto Voyager basemap with verified latitude/longitude coordinates (API Key: cb1_41b1_1_1c25130d48635503dbc1a2a2).

5. Key Facts & Statistical Conclusion Summary

Total Outlets	9,551 restaurants (15 Countries)	Verified comprehensive repository with 0 duplicates.
NCR Dominance	80.3% (7,671 / 9,551 outlets)	New Delhi (5,473), Gurgaon (1,118), Noida (1,080).
True Average Rating	3.44 ★ (Rated only) vs 2.67 ★ (All)	22.5% (2,148) are unrated listings (rating=0.0).
Online Delivery Lift	3.25 ★ vs 2.47 ★ (+0.78★)	Only 25.7% currently offer online delivery.
Table Booking Lift	3.44 ★ vs 2.56 ★ (6.8x votes)	Underused at 12.1%; strong fine-dining demand.
Meal Cost for Two	Median ■400 – ■450 / 46.5% Budget	Luxury outliers reach ■6,000–■8,000 in fine dining.
Top Voted Outliers	Toit (10,934), Truffles (9,667)	Real brand anchors; preserved as genuine signals.

6. Actionable Strategic Recommendations for Zomato

1. Review Incentives for the 22.5% Unrated Merchants: Provide diners with 50 Zomato Pro coins or delivery discounts for submitting reviews for newly listed outlets within 14 days.
2. Target the 74.3% Non-Delivery Outlets: Onboard the top-rated dine-in venues in Delhi and Gurgaon to delivery logistics to drive platform gross merchandise value (GMV).
3. Commercialize Table Reservations: Bundle reservation management and customer queue software into premium merchant packages for Price Range 3 and 4 establishments.
4. Expand Outside Delhi NCR: Deploy focused merchant acquisition initiatives to Bangalore, Mumbai, Pune, and Hyderabad to balance regional portfolio concentration.